

C4_14_COMP4139

UNUK_Report

by Umrah -

Submission date: 12-Dec-2025 02:30PM (UTC+0000)

Submission ID: 270816679

File name: 396585_Umrah_-_C4_14_COMP4139_UNUK_Report_8732205_1469984453.pdf (348.25K)

Word count: 3146

Character count: 18759

MULTIMODAL MACHINE LEARNING FOR PRE-TREATMENT PREDICTION OF PATHOLOGICAL COMPLETE RESPONSE AND RECURRENCE FREE SURVIVAL USING CLINICAL BIOMARKERS AND MRI DERIVED RADIOMICS

*Umrah Zargar, Mohammed Tahir Shaik, Haris Anjum, Kwaku, Mustafa Talat
Department Of Science - Machine Learning (COMP4139 UNUK)
The University of Nottingham*

ABSTRACT

Breast cancer shows large variability with respect to the response of neoadjuvant chemotherapy (NAC), highlighting the importance of early and accurate prediction of pathological complete response (pCR) and recurrence-free survival (RFS). This study proposes a multimodal machine learning model that incorporates 11 biomarkers with 107 features obtained using MRI in the I-SPY 2 dataset, comprising of 400 patients.

Following a step-by-step preprocessing and stage-wise feature selection, ANOVA F-test with $k = 5$ features showed optimal performance for pCR classification, while permutation-based feature selection helped to identify 23 features for RFS regression. The pCR classification model, based on XGBoost with five features, achieved a balanced accuracy of 0.825, while an ensemble model combining Random Forest, Gradient Boosting, and Ridge Regression achieved a mean absolute error of 18.36 for RFS prediction. These findings highlight the complementary predictive value of radiomics and clinical biomarkers and demonstrate the potential of multimodal modelling for individualized NAC treatment support.

Index Terms— Breast cancer, machine learning, neoadjuvant chemotherapy, radiomics, survival analysis.

1. INTRODUCTION

Breast cancer continues to be an important global health issue; it still represents the leading diagnosed malignancy among females and is among the top causes of cancer-associated deaths (Bray et al., 2018). In the United Kingdom, breast cancer similarly tops the list of commonly diagnosed malignancies among females; hence, Neoadjuvant Chemotherapy (NAC) has been established as an important therapeutic modality for those with locally advanced breast cancer. NAC has been routinely used among those diagnosed with locally advanced breast cancer to decrease tumour burden before surgical treatment; however, individual response to tumour reduction shows relatively large inter-patient variability (Tagliafico et al., 2020).

A major clinical endpoint used to evaluate the effectiveness of NAC is pathological complete response (pCR), which refers to the absence of any invasive pathological tumours at

the time of surgery. Patients with pCR show significantly improved survival prognosis with greatly extended recurrence-free survival (RFS) times, while those with residual disease show greatly variable outcomes.

Conventional clinical and histopathological features provide valuable prognostic information; however, they often fail to capture the spatial and biological heterogeneity inherent in breast cancers (Chitalia et al., 2022). Magnetic resonance imaging (MRI) offers high sensitivity for assessing tumour morphology, vascularity, and treatment response, but its interpretation remains largely subjective. Radiomics has emerged as a computational approach that extracts high-dimensional quantitative features related to tumour shape, intensity, and texture, which may encode biological characteristics associated with treatment susceptibility (Tagliafico et al., 2020), (Ye et al., 2020).

Several studies have demonstrated that MRI-based radiomics can successfully model and predict response to NAC in breast cancer patients. Systematic analyses report that radiomics features extracted from pre-treatment MRI scans can identify responders and non-responders with performance comparable to or exceeding that of conventional clinical models, particularly when combined with machine learning classifiers such as logistic regression, random forests, and gradient boosting methods (Granzier et al., 2021).

Unlike many existing studies that focus solely on pCR prediction, this work jointly models both pCR classification and RFS regression within a unified multimodal learning framework. Since there was missing data in the clinical features, this study uses systematic missing data techniques. There were large skewness and large scales of variation among the radiomic features; this pointed to the need for log transformation and normalization. For the pCR classification problem, there was significant imbalance among instances; SMOTE-Tomek resampling was used to achieve balance before model training, while for the RFS regression model it implemented as an ensemble of Random Forest, Gradient Boosting, and Ridge Regression models, indicating strong predictive performance for both tasks.

The overall process flow, including model development for both tasks, is shown in [Figure 1](#).

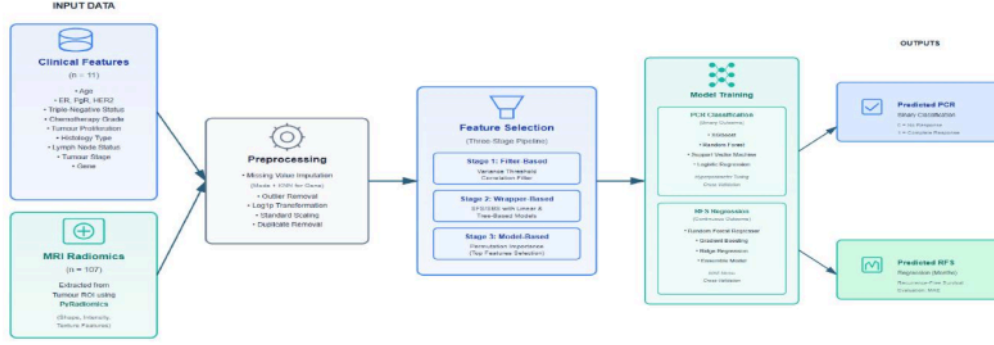


Figure 1: Overview of the multimodal machine learning pipeline for pCR classification and RFS regression.

2. DATA AND METHODOLOGY

2.1. Dataset, Data Pre-processing and Imputation

The training set consisted of 400 patients with 11 clinical biomarkers (such as Age, ER, PgR, HER2, Triple-Negative status, etc) and 107 features obtained using pre-treatment breast MR. The problem set included two target response variables: pCR and RFS. There was a high degree of class imbalance (79% non-pCR, 21% pCR) samples.

Missing values were represented with “999” and then converted to NaN in the clinical features dataset. Gene was found to contain the highest level of missing values; therefore, a missing indicator variable (Gene_Flag) was included before performing KNN-based imputation, as this approach preserves multivariate relationships between clinical features. The median was used for continuous features, and the mode was used for other categorical/ordinal clinical features. Four categorical features (ER, Triple-Negative, Tumour Stage, Gene_Flag) underwent label encoding.

For the radiomic features used in the RFS regression task, the features showed strong skewness; hence, log1p transformation and Z-score scaling were applied to all numeric features. For the pCR classification task, after splitting the data into training and validation sets and applying feature selection techniques, SMOTE-Tomek resampling was applied only to the training set to balance pCR classes. The Tomek-links component was included to remove borderline overlapping samples after oversampling, producing a cleaner decision boundary between pCR and non-pCR classes. For both tasks, the dataset was randomly divided into an 80/20 stratified train-validation split, preserving the original class distribution of the pCR outcome. To further support reproducibility, random seeds were fixed for the train-test split, resampling, and model training procedures.

2.2. Feature Selection

Owing to the high dimensionality inherent within radiomics data, it commonly exhibits the *curse of dimensionality*, whereby the number of derived features tends to exceed the

number of training instances, hence increasing the susceptibility to overfitting (Granzier et al., 2021). It was, therefore necessary to adopt a three-level feature selection method with the specific objective of preserving ER, HER2, and Gene features among others that are clinically important.

2.2.1. Common Filtering and Domain Knowledge Constraints

A filtering step was incorporated before the main analysis to reduce redundancy among the features of the radiomic set. The analysis involved using Pearson’s correlation to identify radiomic features that were largely collinear, and features with correlation coefficients higher than 0.95 were eliminated. The number of radiomic features was reduced from 107 to 82, and this reduced radiomic subset was subsequently used for RFS regression experiments.

2.2.2. Task-Specific Feature Selection

For pCR Classification, ANOVA F-test, Chi-squared test, Random Forest Feature importance, and Recursive Feature Elimination (RFE). ANOVA F-test ($k = 5$) achieved the highest cross-validated balanced accuracy (≈ 0.605) during feature selection, outperforming alternative methods, Figure 2. The two ANOVA-selected features (PgR and missing_Gene) were then combined with the mandatory biomarkers ER, HER2, and Gene features that comprised the final five features, conforming to previously established studies on compact statistical features which generalize best (Granzier et al., 2021).

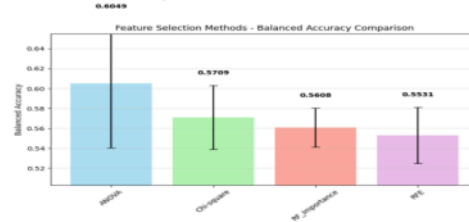


Figure 2: Feature Selection Methods - Balanced Accuracy Comparison

For RFS regression, Sequential Feature Selection (SFS) was applied using Linear, Ridge, and Random Forest models and evaluated using 5-fold cross-validated Mean Absolute Error (MAE). Random Forest-based SFS achieved the lowest MAE

(≈ 19.91), indicating the suitability of tree-based methods for radiomic modelling. However, the final 23-feature subset was obtained using permutation feature importance, combining 20 radiomic features with the three mandatory biomarkers, resulting in an MAE of 18.36 for the ensemble model (Hatamikia et al., 2024).

The final selected feature sets are summarised in *Figure 3*. The PCR model used 5 features, while the RFS model used 23 features, balancing predictive performance, dimensionality, and interpretability (Tagliafico et al., 2020), (Pesapane et al., 2023), (Van Griethuysen et al., 2017).



Figure 3: Cross-validated MAE as a function of the number of selected features using Random Forest-based Sequential Feature Selection for RFS regression

Task	Feature Selection Method	Selection Criterion	Radiomic Features Selected	Mandatory Clinical Features	Total Features
pCR Classification	ANOVA F-test (k = 5)	Cross-validated Balanced Accuracy	2	ER, HER2, Gene	5
RFS Regression	Permutation Feature Importance	Cross-validated MAE	20	ER, HER2, Gene	23

Table 1 Summary of Feature Selection for pCR Classification and RFS Regression

2.3. Model Training and Evaluation

2.3.1 pCR Classification Model Development

The classification study was aimed at predicting pCR using a multimodal set of features that combined radiomic and clinical data. The development of models followed a two-step systematic process, evaluation of initial models on benchmarks and model refinement with feature knowledge. The models were developed and tested using 5-fold Cross Validation with balanced accuracy as the evaluation criterion.

In the first phase, several commonly used classification techniques were tested to determine their performance as baselines. The techniques included Logistic Regression, Random Forest, XGBoost, and Support Vector Machine (SVM) with an RBF kernel. The results demonstrated that with XGBoost achieving the best performance among all models tested. It showed a balanced accuracy of 0.8248 on the test set with high recall indicating good detection of pCR cases.

The second phase involved model refinement using feature selection techniques. Chi-square, Recursive Feature Elimination (RFE), ANOVA F-test, and Random Forest feature importance were tested for selecting the most relevant features. The ANOVA F-test with five features provided the best-balanced accuracy suggesting that a small, optimized feature subset could yield strong classification performance.

A final XGBoost model was trained using the 5-feature multimodal subset which was selected after feature selection. This model achieved the best performance with a balanced accuracy of 0.8248 on the test set and was selected as the final pCR classifier.

2.3.2 RFS Regression Model Development

The regression problem was attempted to predict RFS using the 23-feature set obtained from permutation feature importance together with the three mandatory biomarkers. Development of models was done using a three-step systematic process: evaluation of the initial model on benchmarks and model search with knowledge of features. The models were developed and tested using 5-fold Cross Validation with MAE evaluation criteria.

In the first phase, several commonly used regression techniques were tested to determine how well they could perform as baselines. The techniques included linear regression, ridge regression, Lasso regression, support vector regression, random forest regression, gradient boosting regression, and multi-layer perceptron. Out of all models Random Forest outperformed other models so we experimented with Random Forest too in Sequential Feature Selection.

The second phase involved model development using Sequential Feature Selection (SFS). Three base models: Linear Regression, Ridge Regression, and Random Forest Regression, were used with SFS to identify which features would offer optimal results. It can be observed that among SFS models using Random Forest Regression, Cross-Validated Mean MAE was lowest at 21 features (≈ 19.91) and would suggest that previous research studies carried out in radiomics indicated that models using tree-based techniques achieved superior ability for comprehending intricate relationships prevalent within large amounts of image data (Hatamikia et al., 2024).

The third phase involves the SFS_union set refers to the union of features selected across preliminary SFS runs during baseline comparison. Random Forest Regression achieved the best baseline compared to Gradient Boosting Regression, Support Vector Regression (SVR), and other models. Following this we hypertuned the Random Forest, compared the MAE there's a slight decrease in MAE.

A final ensemble model (combining Random Forest, Gradient Boosting, and Ridge Regression) was subsequently trained using the 23-feature subset derived from permutation feature importance, resulting in a further reduction in

validation error (MAE $\approx$ "18.36"). This was therefore selected as the final RFS predictor and exported with all supporting preprocessing components.

3. Results

Table 2 summarizes the final performance of the multimodal models. The XGBoost classifier achieved a balanced accuracy of 0.825 for pCR prediction using five selected features, while the ensemble regression model achieved an MAE of 18.36 for RFS prediction using 23 features.

Task	Final Model	Features Used	Performance
pCR Classification	XGBoost Classifier	5 features (ANOVA-selected radiomic features + ER, HER2, Gene)	Balanced Accuracy = 0.825
RFS Regression	Ensemble Model	23 features (20 radiomic features + ER, HER2, Gene)	Mean Absolute Error (MAE) = 18.36

Table 2: Final performance of multimodal machine learning models for pCR classification and RFS regression.

Table 3 compares regression models for RFS prediction, where the ensemble approach achieved the lowest MAE, demonstrating improved predictive stability over individual models. *Table 4* reports detailed classification metrics for pCR prediction, showing high recall and balanced precision, indicating robust performance under class imbalance.

Model	MAE
Random Forest	~18.472
Gradient Boosting	~19.535
Ridge Regression	~19.14
Ensemble (Final)	18.36

Table 3: RFS Regression Comparison

Metric	Value
Balanced Accuracy	0.825
Precision	0.77
Recall	0.919
F1-score	0.838

Table 4: pCR Classification Metrics

4. Discussion

Differing from most previous works on pCR prediction, this research addresses response modeling and survival prediction simultaneously within an integrated framework, providing a more comprehensive approach to treatment outcome prediction.(Aerts et al., 2014)

The key strength of the proposed methodology is the rigorous preprocessing pipeline, which included handling missing values systematically, log transformation, standardization, removal of duplicates, and filtering out highly correlated ones. This also decided on the essentiality of high dimensionality and heterogeneity radiomic features. Additionally, clinically relevant biomarkers, namely, ER, HER2, and Gene, were retained throughout the analysis so

that domain knowledge would complement data-driven feature selection.

The multi-stage feature selection strategy further enhanced model robustness. For pCR classification, a compact five-feature subset identified using the ANOVA F-test demonstrated strong discriminative performance, supporting the use of minimal feature sets in moderate-sized datasets. For RFS regression, combining Sequential Feature Selection with permutation feature importance resulted in a 23-feature subset that balanced predictive accuracy with interpretability.

Ensemble models have outperformed linear and kernel-based models on many occasions. An XGBoost classifier yielded a balanced accuracy of about 0.825 in the case of pCR predictions, with robustness to class imbalance and nonlinear interactions between features. Likewise, combining Random Forest, Gradient Boosting, and Ridge Regression attained an MAE of 18.36 for RFS prediction with an improved stability compared to their individual performances.

These encouraging results notwithstanding, some limitations are to be acknowledged: the relatively small size of the dataset (n = 400) may limit generalisability, especially given that radiomic features are high-dimensional.. Thirdly, although class imbalance was addressed with SMOTE-Tomek, the limited number of pCR cases may impact model robustness. Lastly, the absence of external validation limits conclusions regarding clinical deployment

Overall, this study highlights the potential of combining clinical biomarkers and MRI-derived radiomics within a multimodal machine learning framework for predicting breast cancer treatment outcomes. Future studies should enhance robustness and facilitate clinical translation.(Aerts et al., 2014)

5. Conclusion

A successful multimodal machine learning model was developed to predict pCR and RFS in breast cancer patients using a combination of clinical biomarkers and MRI-derived radiomic features.

For the pCR classification task, the XGBoost model achieved a balanced accuracy of approximately 0.825, while the final ensemble model (combining Random Forest, Gradient Boosting, and Ridge Regression) for RFS regression achieved an MAE of 18.36, showcasing the utility of radiomics in improving prognostic predictions.

Our next step should be to gather a wider variety of patient data and incorporate more advanced models for better results. Overall, the study demonstrates the feasibility and effectiveness of multimodal machine learning models in predicting cancer treatment outcomes. Further work is needed to refine these models and explore more advanced techniques, such as deep learning, to achieve even higher accuracy and robustness.

6. REFERENCES

- Aerts, H. J. W. L., Velazquez, E. R., Leijenaar, R. T. H., Parmar, C., Grossmann, P., Cavalho, S., Bussink, J., Monshouwer, R., Haibe-Kains, B., Rietveld, D., Hoebers, F., Rietbergen, M. M., Leemans, C. R., Dekker, A., Quackenbush, J., Gillies, R. J., & Lambin, P. (2014). Decoding tumour phenotype by noninvasive imaging using a quantitative radiomics approach. *Nature Communications*, 5. <https://doi.org/10.1038/ncomms5006>
- Bray, F., Ferlay, J., Soerjomataram, I., Siegel, R. L., Torre, L. A., & Jemal, A. (2018). Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians*, 68(6), 394–424. <https://doi.org/10.3322/caac.21492>
- Chitalia, R., Pati, S., Bhalerao, M., Thakur, S. P., Jahani, N., Belenky, V., McDonald, E. S., Gibbs, J., Newitt, D. C., Hylton, N. M., Kontos, D., & Bakas, S. (2022). Expert tumor annotations and radiomics for locally advanced breast cancer in DCE-MRI for ACRIN 6657/I-SPY1. *Scientific Data*, 9(1). <https://doi.org/10.1038/s41597-022-01555-4>
- Granzier, R. W. Y., Ibrahim, A., Primakov, S. P., Samiei, S., van Nijnatten, T. J. A., de Boer, M., Heuts, E. M., Hulsmans, F. J., Chatterjee, A., Lambin, P., Lobbes, M. B. I., Woodruff, H. C., & Smidt, M. L. (2021). Mri-based radiomics analysis for the pretreatment prediction of pathologic complete tumor response to neoadjuvant systemic therapy in breast cancer patients: A multicenter study. *Cancers*, 13(10). <https://doi.org/10.3390/cancers13102447>
- Hatamikia, S., George, G., Schwarzhans, F., Mahbod, A., & Woitek, R. (2024). Breast MRI radiomics and machine learning-based predictions of response to neoadjuvant chemotherapy – How are they affected by variations in tumor delineation? *Computational and Structural Biotechnology Journal*, 23, 52–63. <https://doi.org/10.1016/j.csbj.2023.11.016>
- Pesapane, F., De Marco, P., Rapino, A., Lombardo, E., Nicosia, L., Tantrige, P., Rotili, A., Bozzini, A. C., Penco, S., Dominelli, V., Trentin, C., Ferrari, F., Farina, M., Meneghetti, L., Latronico, A., Abbate, F., Origgi, D., Carrafiello, G., & Cassano, E. (2023). How Radiomics Can Improve Breast Cancer Diagnosis and Treatment. In *Journal of Clinical Medicine* (Vol. 12, Issue 4). MDPI. <https://doi.org/10.3390/jcm12041372>
- Tagliafico, A. S., Piana, M., Schenone, D., Lai, R., Massone, A. M., & Houssami, N. (2020). Overview of radiomics in breast cancer diagnosis and prognostication. *Breast*, 49, 74–80. <https://doi.org/10.1016/j.breast.2019.10.018>
- Van Griethuysen, J. J. M., Fedorov, A., Parmar, C., Hosny, A., Aucoin, N., Narayan, V., Beets-Tan, R. G. H., Fillion-Robin, J. C., Pieper, S., & Aerts, H. J. W. L. (2017). Computational radiomics system to decode the radiographic phenotype. *Cancer Research*, 77(21), e104–e107. <https://doi.org/10.1158/0008-5472.CAN-17-0339>
- Ye, D. M., Wang, H. T., & Yu, T. (2020). The application of radiomics in breast MRI: A review. In *Technology in Cancer Research and Treatment* (Vol. 19, Issue 1, pp. 1–16). SAGE Publications Inc. <https://doi.org/10.1177/1533033820916191>

Task Weightage &	Data Pre-Processing (10%)	Feature Selection (25%)	ML Method Development (25%)	Method Evaluation (10%)	Report Writing (30%)
Muhammad Tahir Shaik	25	25	30	10	10
Haris Anjum	25	25	30	10	10
Kwaku	20	20	20	30	10
Umrah	15	15	10	25	35
Mustafa Talat	15	15	10	25	35